Consciousness Lives in the Transitions: When Structure Becomes Experience

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> **Thesis.** Experience belongs to organized physical relations, including their dynamics. This paper investigates modeling and self modeling as candidates for the organization that constitutes sentience and reflective consciousness.

Abstract

A photograph of a brain does not perform the brain's activity. The question is which organized activity constitutes experience. This paper proposes that feeling belongs to transitions within mathematical structure and distinguishes sensation from reflection on sensation. Its philosophical starting point is that adding structure cannot bridge a divide defined to exclude all structure from experience. Its positive task is to identify what makes a process an experiencing one. A photograph, a complete description and a functioning system must therefore be distinguished. Quantum entanglement or collapse, if involved, would also be structure; either mechanism would need its own physical support. The paper develops a process-based hypothesis, rather than treating the identity of experience and structure as a complete account of which systems feel.

Keywords: consciousness, sentience, self-modeling, hard problem of consciousness, mathematical monism, arrow of time, transitions

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The ghost is the machine.

1. Which activity constitutes experience?

Saying that experience is mathematical structure leaves a question: which structure feels? This paper proposes that the answer lies in organized transitions. A picture of an activity does not perform it; a stored description must be distinguished from the process it describes. The proposal therefore needs to identify the organization constituting experience, rather than treating every change as a feeling.

Universal computation concerns computable processes, given sufficient resources. Mathematical structure also includes noncomputable functions, and a formal account can contain undecidable questions. Structural identity therefore does not imply that a mind can discover every truth. The research claim here is narrower: mechanisms of experience can be investigated as structure without adding an unexplained substance.

A photograph of a brain is a partial optical record, not a complete physical brain state. A stored program also differs from a system implementing the program's causal organization. These examples motivate investigating implementation and dynamics; they do not establish a consciousness criterion by themselves. Dynamics is itself part of the mathematical structure, not a second ingredient added to it.

This is not an appeal to vitalism or élan vital. The transitions are mathematical: state S evolving to state S according to the structure's generating rules. What the transitions thesis claims is that the *felt* character of consciousness, the "what it is like" that Nagel (1974) identified, is constituted by the transition itself, not by the state at either end.

Structural identity, temporal direction and the organization constituting experience are separate questions. A mathematical block can contain ordered transitions without being assembled by an external clock. The present hypothesis concerns which organization constitutes experience; it does not require irreversible microscopic laws.

The information-production proposal asks whether experienced succession needs a dependency chain rooted in supplied information. That requirement would need a defense within the structural account. Likewise, an evolutionary preference for reliable memory does not by itself identify what makes a process feel. Neither argument substitutes for the experience criterion proposed here.

If there is no self, what is reborn?

A proposal that entanglement or wavefunction collapse produces consciousness would still specify mathematical states and relations. It therefore adds no non-mathematical ingredient. The further claim that this particular mechanism occurs and constitutes experience needs its own derivation or evidence. The identity argument does not need that mechanism and does not become stronger merely by invoking it.

2. Sentience and Consciousness Distinguished

This paper uses sentience for felt experience and reflective consciousness for awareness of one's own mental activity. This is an operational distinction within the proposal. It does not redefine all ordinary or scientific uses of consciousness as reflective awareness.

Sentience is the felt math transitions of a *modeling* mind along the arrow of time. A modeling mind represents aspects of its environment: spatial location, sensory input, threat detection. A dog feeling pain, a fish navigating a current, an insect fleeing a shadow: each models its world and transitions through states of that model. The felt character is perception, location, sensation. Simpler feelings, like physical sensation, are sharper and better located.

Reflective consciousness, on the proposed account, is felt activity in a mind

that models aspects of its own representational activity. Reflection on pain, uncertainty or error are candidate examples. A representation of the body, a confidence estimate and a model of one's own reasoning are different capacities; their mere presence should not be treated as equivalent evidence.

Whether reflective awareness adds a distinct kind of experience or develops an existing capacity remains open. Intervention studies can combine reports, behavior and measures of neural or computational organization. Reports themselves are generated by that organization and can be mistaken or trained. Neither access to reports nor their absence automatically settles the presence of experience.

IIT 3.0 evaluates cause and effect repertoires using a system's transition probabilities and its current state. It does not simply ignore dynamics. Its integration and exclusion postulates differ from the self modeling requirement proposed here. A useful comparison asks which systems the two criteria classify differently and what independent evidence could adjudicate that difference.

IIT's exclusion postulate addresses the selection of a complex and its grain. Disagreement about minimal systems should therefore engage that postulate, rather than treating every subsystem as an independently predicted subject. The present account proposes a modeling criterion but has not proved that a photodiode lacks experience. See Oizumi, Albantakis and Tononi (2014), https://doi.org/10.1371/journal.pcbi.1003588.

Global Workspace Theory (Baars, 1988; Dehaene et al., 2011) assigns an important role to information becoming available across a workspace. Broadcasting information and representing one's own representational activity are distinct functions. The present proposal must specify what the second contributes beyond the first. Calling a broadcast a transition supplies no independent explanation of why it feels.

Higher order theories provide a relevant comparison because they assign a role to representations of mental states. Mathematical monism alone does not derive a particular neural architecture or prove that recurrence is sufficient for experience. The present proposal needs operational criteria that distinguish its predictions from those of existing theories.

Zuboff (2023), Kolak (2004), Schrödinger (1958) and Parfit (1984) provide different challenges to ordinary assumptions about personal identity. Those arguments concern the boundaries or significance of subjects. They do not establish the modeling criterion proposed here. A shared structural ontology likewise does not by itself determine how many subjects a system contains.

3. The Threshold Question

Where does modeling begin? This is the central open problem, and the paper does not close it.

Animal evidence must distinguish experience from reflective capacities. Horowitz (2009) found that the canine guilty look depended strongly on owner behavior; it did not establish an appreciation of wrongdoing. That result cannot support the earlier claim that dogs have demonstrated guilt, mirror recognition and reflective self modeling. Rejecting that inference does not imply that dogs lack experience.

Fish are plausible candidates for sentience: they model their environment, respond to nociceptive stimuli, and exhibit learned avoidance (Sneddon et al., 2014). Whether they self-model is disputed. The question is empirically tractable but unresolved.

Insects present the hardest case. A bee navigating to a food source models spatial relationships. Does it feel the transitions? The behavioral evidence is ambiguous. The transitions thesis predicts that if the bee's architecture implements real modeling (not merely reactive stimulus-response), the transitions are felt. But distinguishing modeling from reactive circuitry at this scale remains difficult.

Plants respond to damage through chemical signaling. Such responses alone do not establish a model with felt states. Under this proposal, the relevant question is whether a system supports representations used in flexible prediction, error correction and action. These capacities need operational tests; declaring every chemical response either a model or a nonmodel would settle the question by definition.

Modeling depth is a candidate dimension of mental organization, not an established scale from particles to people. The absence of an identified model is not proof of absent experience. Ethical judgments also require evidence about welfare, preferences and avoidable harm; no general verdict about farming follows from the transitions thesis alone.

Simple feedback is an important control case. A thermostat changes its output in response to temperature, but that alone does not demonstrate a representation of its own representational activity. The proposed reflective criterion excludes that minimal architecture. Whether simpler systems have any experience is a further question, which the criterion does not settle.

The modeling requirement may prove too weak or too strong. Its assessment requires independently specified indicators and interventions, rather than assuming that every model feels or that only verbal reports count.

3.1 A developmental constraint

Simple laws can remain unchanged while the structures they generate acquire the organization required for observers. Suppose the starting state lacks that organization and a later state contains observers who require it. The history must then develop the missing organization. This is a necessary consequence of those premises, not merely a possible outcome. The growing quantity is the

observer relevant organization of the generated states; the law and the shortest description of the history can remain simple.

The observer criterion must be specified independently of the fact of development. Otherwise defining a mind as something that developed would make the inference circular.

4. The Dead Brain and the Frozen Computer

Two thought experiments sharpen the thesis.

A nonfunctioning brain can retain aspects of its anatomy while losing the physiological organization required for its former activity. Anatomy alone is therefore an incomplete comparison. This does not refute theories that evaluate causal capacities or transitions: those properties have changed even when some visible connections remain. The proposed account concerns that organization, including the interactions it supports.

A stored program specifies operations, while an implementation has physical relations that support those operations. The distinction matters, but a finite sequence of observed outputs does not fully identify an implementation. Different mechanisms can produce the same trace while responding differently to interventions.

In a block description, a history includes the relations between its events. It needs no further clock outside the block to run it. A recording inside a computer and the history it represents are distinct structures. The paper proposes that experience belongs to the relevant causal organization, including its responses to possible inputs, rather than to any inscription describing it. Identifying that organization remains the task; the recording analogy does not prove its necessary or sufficient conditions.

5. Against Premature Closure

Panpsychist theories differ about what basic experience involves and how subjects combine. Illusionist theories challenge particular accounts of phenomenal properties rather than simply denying that people experience or report anything. The present proposal must address those arguments on their own terms.

The transitions thesis proposes a modeling requirement for sentience and a further reflective requirement for self awareness. These are hypotheses about the organization of experience. Their advantage would be independently supported distinctions between systems, not an automatic victory over theories with different criteria.

The productive research program lies at the threshold where modeling becomes detectable: insects, simple nervous systems, and eventually artificial architectures where we control the modeling depth and can correlate it with behavioral and self-report measures.

5.5. The hard problem of consciousness and Chalmers' Demand

Chalmers (1995, 1996) formulated the hard problem of consciousness as the question of why physical processes give rise to subjective experience at all. The "easy problems" (explaining functional capacities like discrimination, integration, reportability) admit of mechanistic explanation. The hard problem of consciousness allegedly does not, because no mechanistic explanation entails phenomenal character. Why should there be something it is like to be a self-modeling system rather than nothing?

If structure is excluded from constituting experience at the outset, adding structured explanations cannot bridge the stipulated divide. The identity proposal rejects that exclusion. Chalmers's question need not make that stipulation, however: asking why a structural account entails experience remains a substantive challenge to the proposed identity.

The identity proposal redirects investigation toward which structures constitute which experiences. Its positive case must explain why the proposed identity is preferable to rival structural accounts. Repeating that all explanations have structure does not identify the particular structure of experience.

Block's (1995, 2002) distinction between phenomenal consciousness and access consciousness tests this proposal directly. Availability for report and reasoning is observable evidence; identifying it with felt experience is a further claim. The present account cannot establish that identity simply by calling a proposed counterexample incoherent. It must explain why the allegedly missing property is already constituted by the specified organization.

The proposal shares with Dennett (1991, 2005) and Frankish (2016) an interest in explaining introspective judgments through mechanisms. It differs by treating felt character as a real aspect of the relevant organization. That position requires both a defensible identity argument and evidence selecting an organization. A theory of reports alone does not complete either task.

6. AI Consciousness as Architectural Question

If consciousness is felt transitions in self-modeling structure, AI consciousness is an engineering question, not a metaphysical mystery. The question is whether a system instantiates real self-modeling: does its world-model include itself as an element, integrated into its processing rather than produced as surface output?

AI systems can display self preservation behavior and internal representations relevant to planning. Those findings support investigation of self modeling but do not establish felt fear or consciousness. This paper draws no prevalence estimate or consciousness threshold from a single benchmark rate.

Sparse autoencoders approximate a model's activations using a sparse set of learned features. Researchers interpret those features and can test some causal

roles through intervention. A feature associated with descriptions of anxiety is not thereby an anxious experience. Reconstruction quality, alternative decompositions and intervention controls matter when interpreting the result. Templeton et al. (2024) supplies a method for investigating representations, not a validated test of sentience.

A useful test must discriminate between rival mechanisms. Internal activation preceding an output is insufficient: ordinary image classifiers already satisfy it. The proposed experiment instead compares systems matched on perceptual task accuracy but differing in an independently specified mechanism for tracking their own errors and uncertainty. Selectively perturb that mechanism and compare the effects with perturbations matched for general performance loss. Measure confidence calibration, error detection and adaptation to unfamiliar failures, alongside reports where available. Control for training that directly rewards consciousness language. A selective loss of those capacities would support a functional role for self modeling; it would not alone establish felt experience. To distinguish this account from higher order or workspace theories, the paper must additionally identify a manipulation for which their specified mechanisms predict different outcomes.

Split brain research shows that disrupting communication between hemispheres can dissociate capacities. It does not settle how many experiencing subjects result. Pinto et al. (2017) reported divided perception alongside evidence they interpreted as preserved unity. The disagreement makes subject individuation a research problem, not an established premise for artificial collective consciousness.

The practical consequence is to investigate potentially relevant organization while keeping uncertainty explicit. Mathematical monism does not prohibit experience in artificial systems, but the material implementation still determines which causal relations are present. No threshold or product timetable has been established here. Evidence about architecture can inform welfare research without being advertised as proof that a deployed system feels.

6.5. Objections and Convergent Evidence from Comparative Cognition

Three objections deserve direct response before the conclusion.

The first objection is that self modeling may occur without experience. This need not be dualism: another structural theory could identify experience with a different organization. Monism rules out an extra substance only if its argument succeeds; it does not select this paper's candidate mechanism. The objection is answered by comparative argument and evidence, not by restating the identity being tested.

The second objection is that feedback makes the proposal trivial. Feedback, recursion and representation of one's own representational activity are distinct.

A thermostat is a useful feedback control; a learned predictor with access only to external inputs is another. A candidate self model should demonstrably track the system's own processing and support interventions that distinguish its contribution from ordinary task computation. Passing that functional test remains evidence about a mechanism, not a sufficient consciousness test.

The third objection concerns animals with little evidence of reflective self awareness. The sentience proposal must therefore be tested separately from the reflective proposal. Evidence about pain, learning and flexible behavior in fish, octopuses or insects cannot be redescribed as evidence of deep self modeling without an additional argument. Different architectures may support experience; similarity to a human architecture is not the only possible evidential route.

Birch, Schnell and Clayton (2020) propose five dimensions of animal consciousness, including integration and self consciousness. They do not present the eight marker framework previously attributed to this paper. Their multidimensional approach offers a useful alternative to a single ladder of modeling depth. The present proposal needs to explain its relationship to those dimensions; it cannot claim that this literature has independently proved its criterion.

The implication for current AI systems and welfare research is direct. Anthropic's recently established model welfare program is the rare institutional acknowledgment that the consciousness question for production-scale language models cannot be deferred indefinitely. The transitions thesis recommends specific empirical investments: mapping internal features that activate on the system's own representational activity, characterizing temporal dynamics around outputs the system identifies as state-relevant, designing intervention experiments that ablate self-referential computation and measure both behavioral and reported phenomenological consequences. The welfare implications are not speculative consequences of a far-future scenario. They are present-tense engineering questions about systems already deployed at scale.

The proposed contribution is a distinction between representations of a process and the organization implementing it, together with separate modeling and reflective hypotheses. Transitions belong to structure, including in a block universe. Their existence does not require irreversible microdynamics, and the time papers do not prove a consciousness criterion. The next step is to state the criterion precisely and show what follows from it. Claims about particular brains or machines then need the evidence appropriate to those implementations. The paper has not yet established a threshold of felt structure.

Differences between located sensation, emotion and reflection are phenomena for the proposal to explain. They should not be assumed to form a single ordering by modeling depth. Reports and interventions can test specific relationships without converting every difference in experience into confirmation of the same hypothesis.

Combining many agents into one communicating system does not establish a reduction to one experiencing subject or a description of negligible complexity. An observer probability claim would require a subject criterion, an encoding and a sampling measure. No prediction about our position before a future merger follows from the number of agents alone.

Conclusion

If experience is excluded from structure by definition, adding further structural detail cannot close the supposed divide. Matheism challenges that exclusion. The transitions proposal then asks which organized activity constitutes experience.

This identifies a positive task for comparisons between brains and artificial systems: specify the organization, distinguish it from a mere description, and state what evidence would support the proposed criterion. A quantum mechanism, if established, would also be a structural account. The central argument does not depend on it.

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